

CatBoost achieved the best test performance: MAPE 4.45%, Adjusted R^2 0.884, MAE 389 MW.

XGBoost followed closely at 5.16% and 0.860. Ridge Regression (10.3%, 0.375) was insufficient for grid operations.

Temperature and hour of day ranked as top predictors in both ensemble models, matching the U-shaped demand curve and intraday peaks from EDA.